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The threat of a majority-minority U.S. alters white Americans' perception of race☆

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ABSTRACT

Racial minorities will soon outnumber white Americans in the U.S. Prior research suggests that this demographic shift is likely to increase white peoples' feelings of threat and anti-minority discrimination. But might this demographic shift also alter *who* is considered a minority in the first place? We tested whether knowledge of an impending "majority-minority" shift in the U.S. would increase threat to white status, leading white perceivers to see mixed-race faces as minorities rather than white—a strategy historically used to preserve white status in the American racial hierarchy. In an initial correlational study, white participants who self-reported greater white status threat perceived mixed-race faces as more Latino than white (Study 1). As compared to those in a control condition, white participants in Studies 2–5 who read about the U.S. demographic shift reported greater white status threat and exhibited reduced perceptual thresholds for categorizing mixed-race faces as Latino, Black, and "not white." A mediation analysis across studies suggests that the status threat white participants experienced from the demographic shift may have lowered their threshold for seeing mixed-race faces as minorities. Our results indicate that the threat of demographic change alters race perception in a manner that increases the number of people who are seen as minorities and who are, therefore, more vulnerable to discrimination.

1. Introduction

The racial demographics of the United States are changing at a rapid pace. Projections indicate that the U.S. will soon be a "majority-minority" nation with racial minority Americans outnumbering non-Hispanic white Americans by 2050 (U.S. Census Bureau, 2018). Although some may view this diversification positively, research suggests that many white Americans are likely to perceive this demographic shift as a threat to their status as the dominant racial group in the United States (Craig & Richeson, 2014a). Awareness of demographic shifts can also cause fears about changing definitions of "Americanness," increased intergroup resource competition, and white "replacement" (e.g., Bai & Federico, 2019, 2021; Danbold & Huo, 2015). Further, white people who are made aware of this impending demographic shift experience more anger and fear toward minorities, express more explicit and implicit anti-outgroup attitudes, show greater support for anti-minority policies, donate more to white than minority recipients, and report a greater willingness to move away from diversifying neighborhoods (Abascal, 2015; Alba, Rumbaut, & Marotz, 2005; Craig & Richeson, 2014b, Craig & Richeson, 2014a; Myers & Levy, 2018; Outten, Schmitt, Miller, & Garcia, 2012; Zou & Cheryan, 2021). Although this work suggests that changing demographics may increase the tendency of white people to exhibit discrimination, it remains unclear how threat from such a demographic shift might influence more basic racial category perception—that is, *who* is seen as a minority in the first place, and thus who will be most vulnerable to amplified discrimination. Here, we

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test whether demographic shifts reduce white perceivers' threshold for seeing a face as belonging to a minority racial group.

One might expect that white perceivers would be motivated to create a larger, more inclusive white category to bolster their numeric majority as a response to demographic changes. However, there is much evidence to suggest that white decision makers may in fact feel a motivation to narrow their perception of "whiteness" in order to preserve white status. For example, the assignment of mixed-race individuals to their "socially subordinate" racial group (i.e., according to hypodescent) has long been associated with the bolstering of white status and the subjugation of minorities (e.g., the notorious "one drop rule"; Banks & Eberhardt, 1998; Hollinger, 2003). Indeed, hypodescent emerged in law in the U.S. to reinforce the high-status of white slavers by protecting the "purity" of whiteness from racial mixing (Davis, 1991). More recently, hypodescent has been associated with the desire among white people to preserve the status quo hierarchy with the white category on top (e.g., Ho, Kteily, & Chen, 2017, Ho, Sidanius, Cuddy, & Banaji, 2013, Krosch, Berntsen, Amodio, Jost and Van Bavel, 2013). This historical and recent empirical evidence suggest hypodescendant race categorization as a powerful tool for maintaining white status. Here, we propose that white perceivers motivated by demographic shifts to bolster their status at the top of the racial hierarchy do so by lowering their perceptual threshold for seeing someone as a minority, thereby keeping the white group small, selective, and high-status.

While it may seem unlikely that race perception could shift at all because such categories are typically regarded as fixed and unchanging (Eberhardt, Dasgupta, & Banaszynski, 2003;), visual processing can change when people are motivated to see things in a preferred way (e.g., Balcetis & Dunning, 2009). Indeed, race perception has been shown to shift as a function of motivational and situational factors (e.g., Freeman & Ambady, 2011) and threat is one such factor that has been identified as an important moderator of race perception (Chang, Krosch, & Cikara, 2016). Aligned with this perspective, white decision makers see mixed-race faces as having more minority features and are more likely to categorize mixed-race faces as minorities when threatened by resource scarcity, physical danger, social exclusion, or an unstable racial hierarchy (Gaither, Pauker, Slepian, & Sommers, 2016; Ho et al., 2013; Krosch & Amodio, 2014; Miller, Maner, & Becker, 2010; Rodeheffer, Hill, & Lord, 2012; Thorstenson, Pazda, Young, & Slepian, 2019). Studies utilizing tools of psychophysiology and neuroscience confirm that threat can alter perceptual thresholds between racial categories and visual processing of race as early as 170 milliseconds (e.g., Krosch & Amodio, 2014, 2019).

Further, recent work demonstrates an effect of demographic change on race ratings and categorization. In one study, Black-white biracial individuals were rated as more "Black American" when they were surrounded by mono-racial[4] Black faces (vs. alone), and this effect was exacerbated by the threat of a waning white majority (Cooley, Brown-Iannuzzi, Brown, & Polikoff, 2018). In another study, the threat of a waning white majority led decision makers to categorize more mono-racial white or mono-racial Latino faces as Latino (Abascal, 2020). However, neither of these studies explicitly test nor rule out alternative explanations despite hypothesizing white status threat as the process through which demographic shifts alter race categorization.

Similarly, these studies suggest demographic change might alter perception of mixed-race faces as minorities, but it remains possible that the demonstrated effects were driven by general shifts in response tendencies or perceived base rate changes rather than shifts in perception. Supporting the plausibility of demographic shift effects on race perception, demographic threats have been shown to alter the own-race bias in face perception (Wilson & Hugenberg, 2010). To our knowledge previous work has not yet examined whether demographic change lowers the perceptual thresholds for seeing mixed-race as minorities, nor has it tested the role of status threat as a process through which this occurs or ruled out plausible alternative accounts for this relationship.

In sum, separate lines of previous research suggest that a) changing U.S. demographics threaten white status and b) that white status can be bolstered by perception of mixed-race individuals as minorities. Thus, we propose that white perceivers are motivated to resolve the status threat they experience from demographic shifts by lowering their perceptual threshold for categorizing mixed-race individuals as minorities, thereby creating a more exclusive white category. In the present research, we examined whether changing U.S. demographics—and the resulting threat to white status—alter the way white Americans visually perceive mixed-race faces. We predicted that white American perceivers would exhibit perceptual thresholds consistent with hypodescent (i.e., over-categorization of mixed-race faces in the "socially subordinate" racial group) and that this would be exacerbated when they were threatened with demographic change.

Because of our primary interest in boundary contraction and visual perception, we focused on a perceptual *Point of Subjective Equality* (PSE) task from psychophysics. This task allowed us to assess the threshold at which faces begin to be seen as minority and to examine the degree of hypodescent (i.e., how far participants deviated from the objective racial content) as well as the amount of "minority content" a face required before it was seen as minority. However, given recent concerns about using morphed stimuli to represent multiracial faces (Chen, Norman, & Nam, 2021; Ma, Kantner, Benitez, & Dunn, 2021) we also extended our work to real faces that were perceived as racially ambiguous.

In an initial study, we examined the correlation between white participants' self-reported status threat and their threshold for categorizing morphed Latino-white faces as Latino.[5] In Study 2, white participants read about the impending "majority-minority" United States (as opposed to a control article about geographic mobility) and we examined their white status threat and perceptual threshold for categorizing morphed Latino-white faces as Latino. We next extended our examination to categorization of non-morphed images of faces that varied in Latino-white racial ambiguity, given recent reports that categorization of morphed and non-morphed stimuli can differ (Ma et al., 2021). Study 3 additionally included both male and female faces to test the generalizability of demographic effects and high vs. low levels of racial ambiguity to bolster our motivational interpretation given race perception should be most susceptible to motivational influences when stimuli are maximally ambiguous (Pauker, Rule, & Ambady, 2010). Next, we examined the effect of changing demographics on white status threat and the perception of mixed Black-white faces as Black (Study 4) to rule out alternative priming accounts. In Study 5, participants had the option to classify faces as "not white," which allowed us to investigate whether changing demographics may produce a more general *minority bias* in the perception of multiracial individuals. Finally, we collapsed across Studies 2, 4, and 5 to examine white status threat as a potential process through which shifting demographics influence white participants' perceptual race thresholds. Supplemental analyses examined non-white participants' perceptual race thresholds and supplemental studies ruled out alternative accounts of shifted base rates and further bolstered the motivated perception account. We report all measures, manipulations,

[4] We use the term "mono-racial" to refer to individuals with parents of the same race and 'mixed-race' to refer to individuals with parents of different races while acknowledging that the racial categories that parents and offspring belong to are socially constructed and that mixed-race individuals are not always racially ambiguous, difficult to categorize, or seen as mixed-race.

[5] We refer to Latino as a racial category (along with Black and white) given a recent push to treat Latino heritage as a race and an ethnicity (Gonzalez-Barrera & Lopez, 2015) but note that Latino identities can be comprised of different races. We use "Latino" when referring to stimuli of known male and female gender identity and "Latinx" when referring to samples with some members identifying outside the gender binary, while acknowledging this term is not universally embraced.

and exclusions in these studies.[6]

2. Study 1

2.1. Method

Ninety-four white-identified (non-Latino, non-Hispanic) Amazon Mechanical Turk workers were recruited in exchange for $1 for 5 min of their time. Twelve participants were removed according to the preregistered criteria from subsequent studies yielding data from 82 participants (mean age = 39.45, *SD* = 13.00; 46 identified as female and 36 identified as male; median household income = $50,000 to $59,999; median education level = High School/GED). Based on a sensitivity power analysis in G*Power, this sample size provided 0.80 power to detect a correlation between white status threat and race categorization threshold of $r = 0.30$ or greater.

After providing informed consent, participants completed a race identification task in which they viewed a series of 95 mixed-race morphed faces that ranged from 100% Latino to 100% white at 10% increments, in a fully randomized order (see Fig. 1A).

Participants classified each face using the P and Q keys, with labels "Latino" and "White" counterbalanced. To create each stimulus face, we combined two unique male "parent" faces and varied the degree to which each parent was represented using morphing software. Selected faces were identified as white or Latino at least 90% of the time from a large database of faces (Ma, Correll, & Wittenbrink, 2015), had neutral expressions, and were matched for facial structure, facial hair, and racial prototypicality. Face images were created to represent each of 11 subcategories ranging from 100% Latino to 0% Latino (i.e., 100% white) at 10% increments of racial ambiguity (e.g., 100% Latino, 90% Latino … 0% Latino). We created the 100% and 0% faces by morphing two Latino and two white "parent" faces, respectively, because morphed faces are judged to be more attractive than non-morphed faces (Langlois & Roggman, 1990). This procedure yielded 95 faces (8–10 per subcategory). Final images were cropped and resized so that the 293 × 400 pixel oval images excluded hairstyles, necks, and ears. All images were then presented on a gray background.

We used a psychophysics approach to probe the effect of changing demographics on the visual processing of race by examining participants' perceptual race thresholds—i.e., we measured the smallest amount of "minority content" a face required to be categorized as a minority. For each participant, the *Point of Subjective Equality* (PSE)—the point on the continuum at which a face was equally likely to be categorized as Latino or white—was calculated by fitting participants' choice data to a cumulative normal function and determining where along the x-axis the resulting curve crossed 50 on the y-axis (as in previous social perception research, e.g., Hackel, Looser, & Van Bavel, 2014; Krosch & Amodio, 2014). A PSE of 50 signifies that a face was perceived in accordance with the racial content of the morphs, with 50% Latino morphs being viewed as Latino 50% of the time. A PSE below 50 indicates that faces were more likely to be perceived as Latino even when they contained less than 50% Latino face content. In other words, participants with a PSE below 50% adopt a lower perceptual threshold for categorizing a face as Latino.

After the face categorization task, participants indicated the degree to which they agreed with the statement that "increases in racial minorities' status will reduce White Americans' status" (from Craig & Richeson, 2014b) on a scale from 0 (strongly disagree) to 100 (strongly agree). Finally, participants provided demographic information and were debriefed.

2.2. Results

We first tested whether white participants were more likely to perceive mixed-race Latino-white faces as Latino than white (i.e., according to hypodescent given the existing racial hierarchy in the U.S.).[7] Although much evidence suggests hypodescent is prevalent when people categorize mixed-race Black-white faces, much less evidence exists for the overcategorization of mixed-race Latino-white faces as Latino (see Young, Sanchez, Pauker, & Gaither, 2020 for a review). Thus, our first goal was to test whether participants' categorization thresholds for mixed Latino-white faces indicated a pattern of hypodescent. To the extent that racially ambiguous mixed Latino-white faces are seen as more Latino than white, mean PSE should be lower than 50. Indeed, a one-sample *t*-test revealed that mean PSE ($M = 46.11$, $SD = 13.55$) was significantly below 50, $t(81) = 2.60$, $p = .01$, 95% CI = [43.13, 49.08], consistent with the phenomenon of hypodescent for Latino faces (Fig. 2A). That is, on average, faces required less than 50% Latino content to be categorized as Latino.

To test our main hypothesis, we next examined the relationship between participants' perceptions of racial minority status as a threat to white Americans' status and their point of subjective equality scores. As expected, participants who indicated that white status is threatened by minority groups' gains had a lower threshold for perceiving mixed race faces as Latino, $r(80) = -0.30$, $p = .006$, $t = 2.82$, 95% CI = [−0.49, −0.09] (Fig. 2B).

2.3. Discussion

Study 1 provided an initial test of our hypothesis that status threat among white perceivers is associated with greater perception of mixed-race faces as minorities. Consistent with this hypothesis, white participants who believe that minority advancement undermines white status had a lower threshold for seeing a mixed-race face as Latino. In Study 2, we aimed to manipulate white status threat via shifting demographics to examine their causal effect on race perception.

3. Study 2

Prior work has shown that white participants who read about the coming "majority-minority" demographic shift tend to perceive minority status gains as synonymous with white status losses (Craig & Richeson, 2014a). Thus, in Study 2 we used this manipulation to induce status threat among white participants. We predicted that participants who read about the impending "majority-minority" shift in the U.S. would feel greater status threat and would indicate a lower threshold for categorizing mixed-race faces as Latino.

3.1. Method

1118 white-identified (non-Latinx, non-Hispanic) Amazon

[6] Exclusions in all studies were made according to the criteria of the preregistered Studies 2, 4, and 5 (i.e., the intention to exclude participants who incorrectly categorized more than half of the 100% race faces, failed to complete more than 90% of the categorization trials, or reported they did not faithfully answer questions in our study). Although we originally preregistered an intent to include all non-Black and non-Latino/Hispanic-identified participants, we restrict present analyses to white-identified participants to examine the role of white status threat more clearly. We did not preregister, but present results that exclude repeat responders and participants with race categorization threshold scores beyond +/− 3SD (though these exclusions do not change the pattern or significance of results - see "Analysis with Only Preregistered Exclusions" in the supplement for results without these non-preregistered exclusions). Exclusions were distributed evenly across threat conditions in all studies, χ^2's < 0.57 *p*'s > .45. We present the results of two-tailed *t*-tests throughout and define significance as a *p*-value < .05 or a CI that does not contain zero.

[7] To ease comprehension, we refer to "faces" throughout the manuscript. However, we examined only male faces in Studies 1, 2, 4, and 5, limiting the gender generalizability of our effects in these studies.

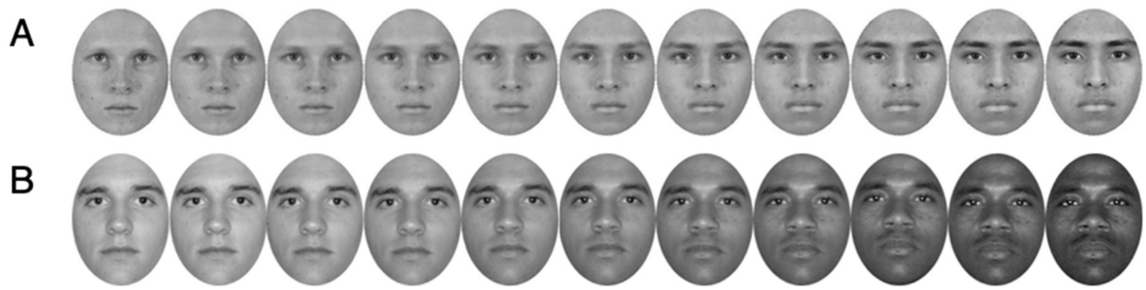

Fig. 1. Study stimuli. Sample morph continuum of faces ranging from (A) 100% white to 100% Latino used in Studies 1 and 2 and (B) 100% white to 100% Black used in Studies 4 and 5. A single face from each continuum was used on each trial.

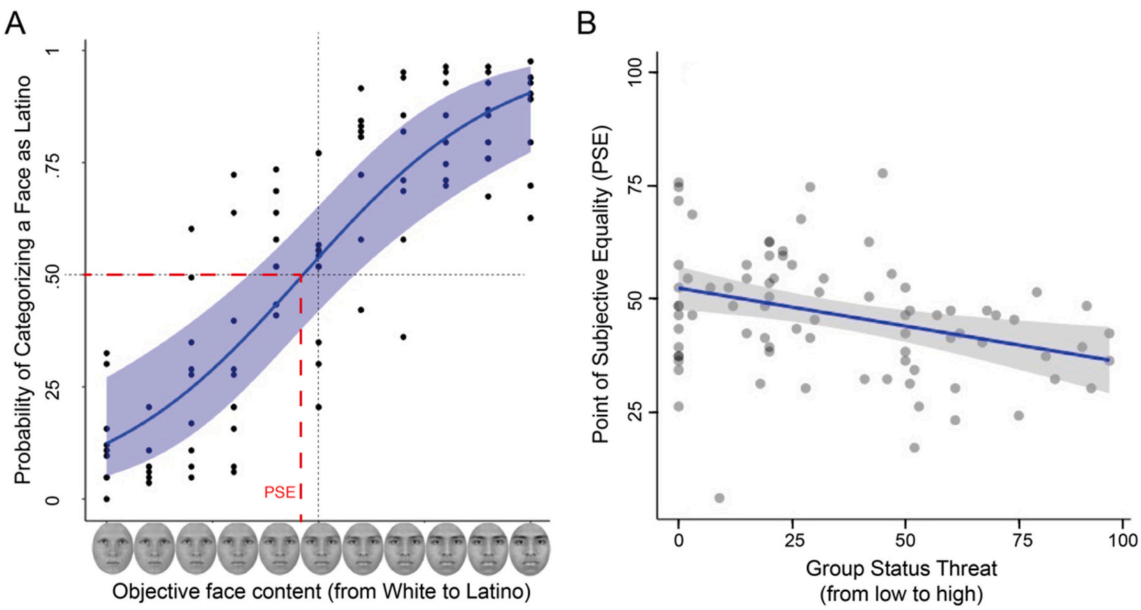

Fig. 2. Study 1 results. (A) The curved line represents the estimated probability of categorizing a mixed Latino-white face as Latino as a function of the racial content of face morphs (shaded region = standard error; dots represent each of the 95 faces). (B) Continuous negative relationship between white status threat and point of subjective equality (PSE). Solid line represents the regression line and shaded region = standard error.

Mechanical Turk workers were recruited to participate in this study in exchange for \$0.16/min of their time. Excluding participants based on our preregistered and reported exclusion criteria yielded data from 935 participants (mean age = 38.54, SD = 15.69; 553 identified as female and 376 identified as male, 2 identified as non-binary, and 4 did not report their gender identity; median household income: \$60,000 to 69,999; median education level: 4-year college degree). Based on a sensitivity power analysis in G*Power, this sample size provides 0.80 power to detect an effect of threat condition on PSE of effect size Cohen's d = 0.18 or greater.

After providing informed consent, participants were randomly assigned to either the threat condition or the control condition. In the threat condition, participants read about the projection that racial minority groups will make up a majority of the U.S. population by 2042. Those in the control condition read about the growth of geographic mobility in the U.S. (as in Craig & Richeson, 2014a).[8] Next, participants answered three comprehension questions about the article they read (see supplement for all materials and measures). Participants then completed the same race categorization task and white status threat measure from Study 1, provided demographic information, and were debriefed.

Study 2 was originally run as a preregistered pilot (S2a) in which we noted our intent to collect at least 400 participants to achieve 95% power to detect a small-to-medium effect size (Cohen's d = 0.35). We were unable to reach our preregistered sample size in this first wave and while the effect the demographic manipulation on PSE was in the predicted direction with n = 294, it was non-significant, p = .20. Thus, we next ran a higher-powered replication study (S2b, n = 641) which yielded significant results, p = .02 (see supplement for full separate results). Adding a sample variable (S2a vs. S2b) and its interaction with the manipulation to the model did not improve fit, F = 0.05, p = .95, nor was there a significant main effect of sample or interaction of sample and manipulation on PSE (p's > .77). Thus, we felt confident merging the samples and present the combined results to obtain a more stable effect size estimate. The pilot Study 2a included measures of ideology and zero-sum beliefs, the Attitudes toward Blacks scale, the Ascent of Man scale, and a feeling thermometer (Amodio & Devine, 2006; Brigham, 1993; Craig & Richeson, 2014b; Krosch & Amodio, 2014; Kteily, Bruneau, Waytz, & Cotterill, 2015) to test exploratory alternative accounts for how this manipulation might operate on race perception. However, the manipulation failed to influence these measures (p's > .11; except the ascent of man scale, in which the threat condition increased humanization across all races, ps < .05; see supplement), so we omitted

[8] Demographic projections have changed only very slightly since these stimuli were developed (i.e., the nation is now projected to become "majority minority" by 2050 rather than 2042) so we deemed the original stimuli appropriate for the current investigations.

them from the replication study (S2b).

3.2. Results

As in Study 1, a one-sample *t*-test revealed that mean PSE (M = 47.20, SD = 14.10) was significantly below 50, $t(934) = 6.06$, $p < .001$, 95%CI = [46.29, 48.11]. This result is consistent with the phenomenon of hypodescent for mixed-race Latino-white faces. Next, we submitted participants' white status threat measure to an independent t-test and found that participants in the threat condition reported greater threat to white status ($n = 474$, $M = 43.47$, $SD = 27.94$) than those in the control condition ($n = 461$, $M = 31.91$, $SD = 27.94$), $t(933) = 6.18$, $p < .001$, Cohen's $d = 0.40$, 95%CI = [0.27, 0.54]. Finally, we examined the effect of the demographic manipulation on racial categorization thresholds with an independent t-test. As expected, there was a significant effect of condition such that participants in the threat condition exhibited a lower PSE (M = 45.99, SD = 13.41) than those in the control condition (M = 48.55, SD = 14.69; See Fig. 3A), $t(933) = 2.69$, $p = .007$, Cohen's d = 0.18, 95%CI = [0.05, 0.30].[9]

3.3. Discussion

White participants who read about a coming "majority-minority" in the U.S. had a lower threshold for seeing mixed race faces as Latino (i.e., faces required less Latino content to be categorized as Latino in the threat condition). The results demonstrate that mixed-race Latino-white faces are categorized according to their "socially subordinate" race (i.e., according to hypodescent), replicate findings that demographic changes induce greater white status threat (Craig & Richeson, 2014b), and provide the first causal evidence that demographic changes may influence white perceivers' perceptual threshold for seeing mixed Latino-white faces as Latino. Taken together, Studies 1 and 2 provided initial support for our hypothesis that white status threat may account for perceptual changes in race categorization that result from demographic shifts. In Study 1, we demonstrated that greater white status threat was associated with lower thresholds for seeing mixed-race faces as minorities. In Study 2, we found the impending demographic shift induced greater white status threat and lower thresholds for seeing mixed-race faces as minorities. We also ruled out alternative mediator candidates that were unaffected by the demographic manipulation (in Study 2a). In Study 3, we used a similar paradigm that was intended to address potential limitations to the generalizability of our findings.

4. Study 3

The results of Study 2 suggest that threats from demographic shifts lower perceptual thresholds for seeing mixed-race faces as members of a minority group. However, our reliance on computerized morphs of real faces left open the possibility that the effect we observed could be limited to morphed faces. In Study 2, we also limited our stimuli to male faces because research has demonstrated clearer patterns of racial bias toward males in general (Sidanius & Veniegas, 2013) and in hypodescent studies specifically (Ho, Sidanius, Levin, & Banaji, 2011). Thus, it remained possible that the threat of a waning white majority might have a different effect on perception of female faces (see Abascal, 2020). To address these issues, we conducted a conceptual replication using male and female faces from a database of non-morphed faces (Ma et al., 2015) which allowed us to capitalize on the noted benefits of using real faces in tandem with computer-generated faces (see Chen et al., 2021; Ma et al., 2021).

In Study 2, we used face stimuli carefully titrated to increase parametrically in racial content because of our primary interest in examining boundary contraction and visual perception. However, this method precluded us from easily testing whether demographic effects on race perception are restricted to the most maximally ambiguous faces. In the current study, we used real faces judged by others to be maximally ambiguous (i.e., categorized as white and Latino equally) or only somewhat ambiguous faces (i.e., categorized mostly as white but occasionally as Latino) by a U.S. rater sample (Ma et al., 2015). Given strong hypodescent effects observed in Studies 1 and 2, we omitted faces that were already mostly categorized as Latino, reasoning that there would be little opportunity for threat to increase minority categorization of these faces. Finally, to further rule out alternative explanations for the effect of demographic threat on race categorization, we also included additional measures that previous research has examined in relation to demographic shifts.

We predicted the demographic manipulation would increase the likelihood of categorizing faces as Latino, conceptually replicating Study 2 with real faces. We further predicted that highly ambiguous faces would show the strongest effect of demographic condition given perception of social categories is particularly susceptible to the influence of motivations when stimuli are maximally ambiguous (e.g., Pauker et al., 2010) and also aimed to test the exploratory hypothesis that less ambiguous faces would show a weaker effect or no effect of demographic condition. Although we included both male and female faces in order to broadly examine the influence of demographic shifts on categorization, we remained agnostic about this interaction given the mixed findings in the literature about white participants' use of hypodescent for female and male faces (Abascal, 2020; Ho et al., 2011).

4.1. Method

191 white-identified (non-Latinx, non-Hispanic) Amazon Mechanical Turk workers were recruited in exchange for $0.16/min of their time (Mean age = 34.32, SD = 13.38; 112 identified as female, 76 identified as male, one as agender, one as nonbinary, and one as "other"; median household income of $50,000 to 59,999).

The procedure for Study 3 was identical to Study 2 with a few exceptions. After the demographic shift manipulation, all participants saw the faces of ten real individuals selected from the Chicago Face Database (CFD; Ma et al., 2015) rather than face morphs. Faces were selected based on their self-identification in the CFD as Latino or white and their subjective race categorization rates from a U.S. rater sample (ibid.). Five faces were selected based on their "high ambiguity," with each face having been almost evenly categorized as white and Latino (on average 43% and 44% of the time, respectively). Five other faces were chosen based on their lower level of ambiguity, each having been categorized as White ~70% of the time and Latino ~22% of the time (see Fig. 4).

Participants categorized faces based on the response options from Ma et al. (2015), i.e., Asian, Black, Hispanic/Latino, White, or Other, plus an additional "Biracial/Multiracial" category. For consistency with our other studies, we coded "White" categorization as white and "Biracial/Multiracial", "Asian", "Other", "Hispanic/Latino", and "Black" as not white (data are available on OSF for more granular analysis).

To estimate an approximate effect size and sample size for the present study, we restricted the data from Study 2 data to the 50/50 and 70/30 white-Latino faces and regressed categorization choices onto these two ambiguity levels, demographic condition, and their interaction. This yielded a demographic condition effect size of log odds = 0.56, corresponding with Cohen's d = 0.31, confirming our suspicion that demographic threat effects would be stronger on a limited subset of maximally ambiguous stimuli than on threshold (i.e., Cohen's $d = 0.17$ in Study 2). This analysis also yielded an interaction effect of log odds = 0.59. An a priori power analysis using the R package SIMR with the present design suggested 200 participants would provide at least 90% power to detect effects of this magnitude in a new sample. A sensitivity power analysis revealed that our achieved sample size provided 0.80

[9] This p-value exceeds the adjusted alpha of .038, calculated using the GroupSeq R package according to Lakens (2014) in order to reduce the chance of Type I error due to sequential analysis.

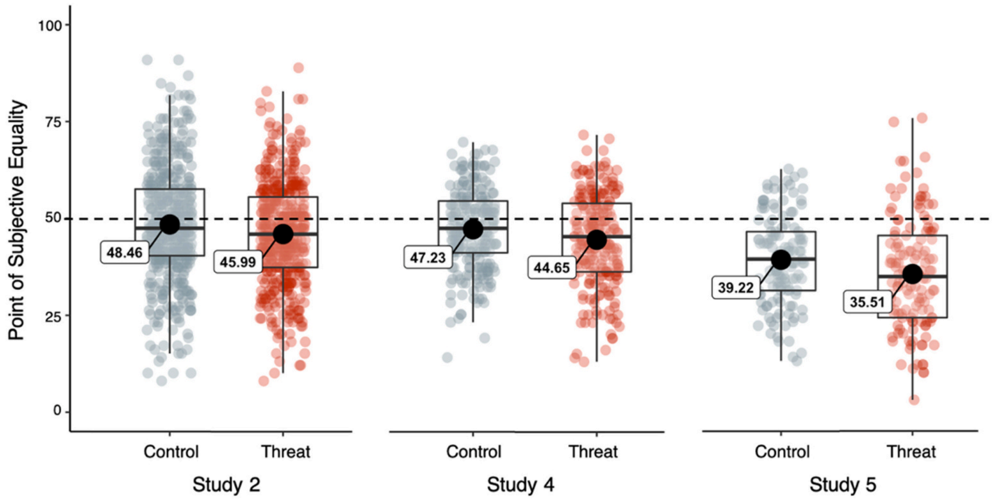

Fig. 3. Results from Studies 2, 4, and 5: White participants in the threat condition (vs. control) had a lower threshold for categorizing mixed-race faces as Latino (Study 2), Black (Study 4), or "not white" (Study 5). Each dot represents a single participant's PSE score (jittered horizontally), boxes represent the 1st and 3rd quartiles split by the median, and whiskers represent 95% confidence intervals (Patil, 2021). All mean PSE values were significantly below 50, p's $< .03$, suggesting perceptual hypodescent was present in both conditions but was heightened in the threat condition.

Fig. 4. Sample stimuli used in Study 2. Top row (high ambiguity) faces were listed in the Chicago Face Database as having been categorized as white and Latino approximately equally; Bottom row (low ambiguity) faces were listed as having been categorized as white about three times as often as Latino. Participants categorized all faces in a random order.

power to detect a main effect of demographic condition on race categorization of log odds = 0.60 or greater and an interaction between demographic condition and ambiguity-level on race categorization of log odds = 0.58 or greater.[10]

After the face categorization task, participants completed the same white status threat measure used in Studies 1 and 2 and indicated the degree of threat they felt in the areas of diversity, prototypicality, and uncertainty, as well as realistic, symbolic, and system threats that previous research has investigated in relation to demographic shift manipulations (Danbold & Huo, 2015; Outten, Lee, Costa-Lopes, Schmitt, & Vala, 2018; see Supplement). Participants then reported how likely they believe it is that non-white people will outnumber white people in the U.S. by 2042. These measures were presented in a randomized order. Finally, they reported their demographic information.

4.2. Results

Reading about the demographic shift significantly influenced participants' belief that non-white people will outnumber white people in the U.S. by 2042: Participants in the threat condition believed a U.S. Majority-Minority was more likely to occur (n = 97, M = 5.65, SD = 1.22) than participants in the control condition (n = 94, M = 4.86, SD =

[10] Because the model used to reanalyze Study 2 data and to analyze Study 3 data was a mixed-effects logistic regression, coefficients express log odds.

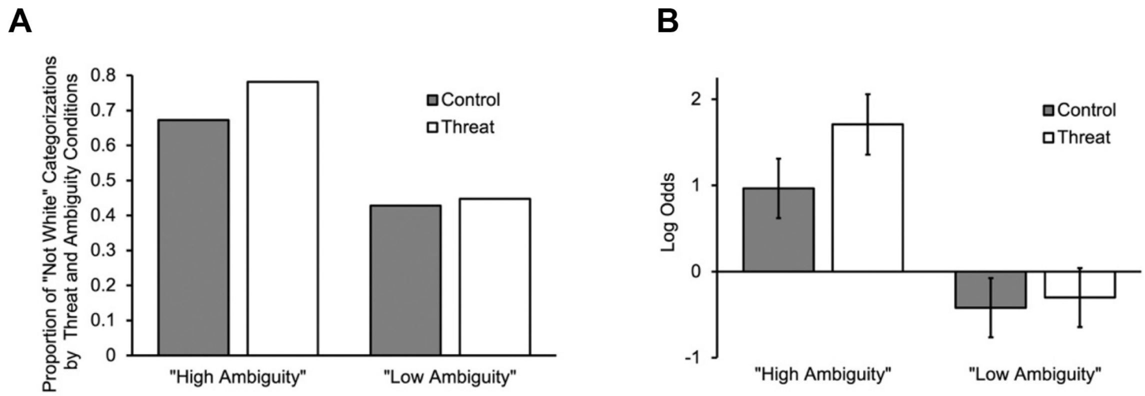

Fig. 5. The effect of threat and ambiguity conditions on (a) proportion of "not white" categorization and (b) log odds of categorizing faces as "not white" ± standard error of coefficients, as estimated the from multilevel logistic regression.

1.56), $t(185) = 3.89$, $p < .001$, Cohen's $d = 0.57$, 95%CI = [0.27, 0.86]. As in Study 2, participants in the threat condition ($M = 3.11$, $SD = 1.66$) reported greater White status threat than control participants ($M = 2.59$, $SD = 1.87$), $t(185) = 2.06$, $p = .04$, Cohen's $d = 0.30$, 95%CI = [0.01, 0.59]. Informing participants about the demographic shift had no significant effect on any of the other measures (all p's > 0.49; see Table S1 in the supplement).

Next, to test whether the threat of demographic change predicted race categorization of real faces, we regressed face categorization (coded as white or not white) onto threat condition, ambiguity, and their interaction with random effects for participants and individual face stimuli (following Judd, Westfall, & Kenny, 2012) using the glmer function from R package lme4. Replicating the findings of Study 2 using non-morphed faces, we found a significant main effect of threat, log odds = 0.74, SE = 0.26, $z = 2.87$, $p = .004$, such that the tendency to categorize faces as not white was greater for participants in the threat condition than the control condition. Furthermore, the main effect of ambiguity also emerged, log odds = −1.38, SE = 0.45, $z = -3.10$, $p = .002$, such that ambiguous faces were more likely to be categorized as not white than less ambiguous faces. These main effects were qualified by the predicted threat x ambiguity interaction, log odds = −0.62, SE = 0.23, $z = -2.69$, $p = .007$. Follow-up comparisons revealed that ambiguous faces in the threat condition were more likely to be categorized as not white than those in the control condition (log odds = −0.74, SE = 0.26, $z = -2.87$, $p = .004$). However, there was no significant difference between the threat and control conditions in the degree to which the less ambiguous faces were categorized as not white (log odds = −0.12, SE = 0.25, $z = -0.48$, $p = .63$; See Fig. 5.

In other words, 78% of highly ambiguous faces were categorized as not white in the threat condition compared to 67% in the control condition, $\chi^2 = 13.80$, $p < .001$. Of the low ambiguity faces, 45% and 43% were categorized as not white in the threat and control conditions, respectively, and these proportions did not significantly differ, $\chi^2 = 0.30$, $p = .58$. Furthermore, adding gender to the model did not improve fit ($\chi^2 = 0.93$, $p = .63$) and it did not predict race categorization independently or interactively with condition or ambiguity, b's < 0.45, p's $> .46$.

4.3. Discussion

Study 3 conceptually replicated and extended the results of Study 2. White participants who read about the demographic shift were more likely to perceive real "racially ambiguous" Latino-identified and white-identified people as Latino than white. This effect was strongest for more racially ambiguous faces, a result that is consistent with previous research showing that motivation typically exerts the greatest influence on visual perception of ambiguous stimuli (e.g., Pauker et al., 2010). Further, the demographic manipulation affected white status threat, but did not affect threats related to diversity, prototypicality, and uncertainty, or realistic, symbolic, and system threats. This particular result rules out these additional alternative mediator candidates while simultaneously supporting the notion that the impending demographic shift increases status threat for white people. All together, these findings provide further support for the hypothesis that white status threat from demographic changes influence who is perceptually categorized as a minority. It also rules out the possibility that the effects of demographic shifts on race perception that we found in earlier studies rely specifically on computer-morphed images.

However, it is possible that the manipulation used in Studies 2 and 3 included content that could generally bias participants toward Latino categorization *independent* of white status threat. That is, participants read statements like "minorities [in America] are increasing…largely boosted by a surge in Hispanic births." Although "Hispanic" typically refers to people from or descended from Spanish-speaking populations and "Latino" typically refers to people from or descended from Latin American populations, the two are often colloquially interchangeable. Thus, it remained possible that our manipulation influenced perception of mixed-race faces because it primed a perceptual readiness to see Latino features, rather than motivating white participants to use hypodescendant categorization to protect white status as we have proposed. Study 4 to was run to address this possibility.

5. Study 4

In Study 4, we sought to replicate the effects of Study 2 using non-Latino facial stimuli to provide additional evidence for our hypothesis that demographic shifts alter perception because white perceivers are motivated to draw a narrower perceptual boundary around their own race when experiencing status threat. To do this, we examined the effect of demographic shifts on the perception of Black-white mixed-race individuals as Black, a racial category that was mentioned in our demographic manipulation but not implicated as contributing to changing demographics or as increasing in population. If the effect of a coming majority-minority U.S. on race perception is driven by the motivation to protect white status, then the demographic shift manipulation we used (which implicates Latino growth) should increase the perception of any mixed-race individual as a minority rather than white. This should be the case even for mixed-race individuals who are not Latino, such as those individuals who are Black and white. If, on the other hand, the effects of Studies 2 and 3 were driven by priming Latino identities or tuning participants to Latino features, this manipulation should have little effect on perception of mixed Black and white individuals.

5.1. Method

484 white-identified (non-Latinx, non-Hispanic) Amazon Mechanical Turk workers were recruited in exchange for $0.16/min of their time. Excluding participants based on our preregistered and reported exclusion criteria yielded data from 455 participants (mean age = 38.27, *SD* = 12.59; 259 identified as female, 195 identified as male, and 1 did not provide their gender identity; median household income: $50,000 to 59,999; median education level: 4-year College Degree). We preregistered our intent to collect at least 400 participants to achieve 95% power to detect a small-to-medium effect size (Cohen's $d = 0.35$).[11] A G*Power sensitivity analysis revealed our achieved sample size of 455 provides 0.80 power to detect an effect size of d = 0.26. This study was identical to Study 2 except participants saw individual faces from a series of Black and white face morphs and categorized them as "Black" or "White" (see Fig. 1B).

5.2. Results

As in Studies 2 and 3, a one-sample *t*-test revealed that mean PSE ($M = 45.91$, $SD = 11.31$) was significantly below 50, $t(454) = 7.70$, $p < .001$, 95%CI = [44.87, 46.67], consistent with the phenomenon of hypodescent for Black-white faces. Once again, participants in the threat condition ($n = 231$, $M = 44.28$, $SD = 30.09$) reported greater white status threat than those in the control condition ($n = 224$, $M = 32.37$, $SD = 28.57$), $t(453) = 4.21$, $p < .001$, Cohen's $d = 0.41$, 95%CI = [0.22, 0.59]. Consistent with our hypothesis, we found a significant effect of condition such that participants in the threat condition exhibited a lower PSE ($M = 44.64$, $SD = 11.93$) than those in the control condition ($M = 47.23$, $SD = 10.48$; See Fig. 3B), $t(453) = 2.56$, $p = .01$, Cohen's $d = 0.23$, 95%CI = [0.05, 0.42].[12]

5.3. Discussion

Participants in the threat condition read about a coming "majority-minority" in the U.S. driven by Hispanic growth but nonetheless exhibited a lower threshold for seeing mixed-race faces as Black than those in the control condition (i.e., faces needed to contain less Black content to be categorized as Black in the threat condition). This suggests that our previous results were not likely caused by beliefs about a greater number of Latinos in the population or priming participants to see Latino features. Rather, these results support the account that the threat of changing demographic motivates white perceivers to protect their status as the dominant racial group by raising their perceptual threshold for who qualifies as white. This finding extends previous work suggesting that hypodescent in race categorization is used as a tool for maintaining the racial hierarchy with white on top (e.g., Ho et al., 2017; Krosch, Berntsen, Amodio, Jost and Van Bavel, 2013) by showing that this same tendency can be provoked by white status threat due to demographic changes.

In this study, we used a task that forced binary categorization of Black-white mixed-race faces into the mono-racial categories of their "parent" faces, Black and white. However, a growing body of recent research suggests that such Black-white mixed-race faces are often perceived as members of other racial minority groups (e.g., Middle Eastern) and that giving perceivers more expansive categorization options is more ecologically valid (e.g., Chen, Pauker, Gaither, Hamilton, & Sherman, 2018; Feliciano, 2016; Nicolas, Skinner and Dickter, 2019). Thus, the extent of categorization of mixed-race individuals as minorities may be underestimated if the response options are too limited to fully capture participants' spontaneous perceptions. To replicate the effects of changing demographic on race categorization thresholds with a more ecologically valid choice task, and to get closer to true proportions of white vs. minority categorizations under threat vs. control, we conducted Study 5.

6. Study 5

In Study 5, we examined the effect of demographic shifts on the perception of Black-white mixed-race individuals as "White" or "Not White." These labels gave participants more expansive choice options for the mixed-race Black-white faces that they may have perceived as neither white nor Black and they align nicely with findings that perceivers tend to spontaneously assess novel individuals as "white or not" upon initial encounter (e.g., Chen et al., 2018; Stroessner, 1996; Zarate & Smith, 1990). Importantly, they also provide the binary choice data needed to fit race perception thresholds. We predicted once again that participants who read about the demographic shift would exhibit a lower PSE.

6.1. Method

289 white-identified (non-Latinx, non-Hispanic) Amazon Mechanical Turk workers were recruited to participate in this study in exchange for $1 for approximately 6 min of their time. We preregistered our intent to collect 400 participants but failed to reach this number in a timely fashion given our exclusions of MTurkers who had previously participated in one of the previous studies. However, we assumed the effects observed in this study could be larger than previous studies due to the inclusion of the broader "Not White" categorization option, so we proceeded with analysis. Excluding participants based on our preregistered exclusion criteria yielded data from 263 participants (mean age = 37.07, *SD* = 11.53; 140 identified as female, 121 as male, one as a transman, and one did not report their gender identity; median household income: $50,000 to 59,999; median education level: 4-year College Degree). A sensitivity analysis in G*Power revealed our sample size provided 0.80 power to detect an effect size of d = 0.35. The procedure and materials were identical to Study 4 except that participants categorized faces as "White" or "Not White."

6.2. Results

As in previous experiments, mean PSE ($M = 37.35$, $SD = 13.19$) was significantly below 50, $t(262) = 15.55$, $p < .001$, 95%CI = [35.75, 38.96] and participants in the threat condition ($n = 132$, $M = 48.16$, $SD = 31.55$) reported greater white status threat than those in the control condition ($n = 131$, $M = 34.89$, $SD = 28.76$), $t(261) = 3.56$, $p < .001$, Cohen's $d = 0.44$, 95%CI = [0.19, 0.69]. Critically, we also replicated the significant effect of condition on race categorization threshold from previous studies. Participants in the threat condition exhibited a lower PSE ($M = 35.51$, $SD = 14.44$) than those in the control condition ($M = 39.22$, $SD = 11.56$; See Fig. 3C), $t(261) = 2.30$, $p = .02$, Cohen's $d = 0.28$, 95%CI = [0.04, 0.53]. That is, participants who read about a coming "majority-minority" in the U.S. had a lower threshold for seeing mixed-race faces as "Not White" (i.e., faces needed to contain less Black content to be categorized as "Not White" in the threat condition).

[11] We include in this study the data from a small-N pilot study (S4a) which yielded non-significant results in the expected direction, in combination with our preregistered replication study (S4b), which yielded significant results. S4a was identical to S4b except that it included the alternative mediator candidates from Study 2a which did not show any significant effects of the manipulation, so we omitted them from Study 4b. Adding a sample variable (S4a vs. S4b) and its interaction with the manipulation to the model did not improve fit, $F = 0.21$, $p = .81$, nor was there a significant main effect of sample or interaction of sample and manipulation on PSE (p's $> .62$). Thus, we felt confident merging the samples and present the combined results to obtain a more stable effect size estimate and to empty our file drawer (see supplement for separate results and alternative mediator analyses).

[12] This *p*-value exceeds the adjusted alpha of .03, calculated using the GroupSeq R package according to Lakens (2014) in order to reduce the chance of Type I error due to sequential analysis.

6.3. Discussion

In Study 5, participants who read about a coming majority-minority demographic shift had a lower threshold for seeing mixed Black-White faces as minorities, once again replicating the effect of changing demographic on race perception. By using a more expansive "Not White" label, we also revealed that white participants required mixed-race faces to only have ~37% Black content to be categorized as a "Not White" (compared to 46% in Study 4 which used identical stimuli and manipulation). This suggests that traditional "Black" vs. "White" labels may have limited past studies' ability to detect the true magnitude of hypodescent in race categorization (as suggested by previous research; Chen et al., 2018; Feliciano, 2016; Nicolas, Skinner, & Dickter, 2019). The effect of the status threat manipulation on PSE was also descriptively larger in this study (d = 0.28) than in Study 4 (d = 0.23), which suggests that white perceivers may draw more exclusive boundary around whiteness when they have greater flexibility to categorize mixed race faces as non-white, though these did not differ significantly (see supplement for face stimuli and response option moderator analyses).

7. Data across race threshold experiments

To establish a more stable estimate of effect size and to more formally test whether white status threat mediates the effect of demographic shifts on race perception thresholds, we collapsed across all experimental data that included the point of subjective equality outcome measure (Studies 2, 4, and 5).

7.1. Mini meta-analysis

First, we performed an internal mini meta-analysis (Goh, Hall, & Rosenthal, 2016) using a fixed effects analysis in which the mean effect size (i.e., Cohen's d) was weighted by sample size. As expected, this revealed an overall significant effect of condition on PSE, mean d = 0.24, $SE = 0.05$, Z = 4.80, $p < .001$, 95%CI = [0.14, 0.33], such that participants in the threat condition exhibited a lower threshold for categorizing a mixed-race face as a minority (See Fig. 6).

7.2. Test of white status threat as a mediator

Although the individual studies examining the demographic manipulation on PSE were not powered to detect the presence of mediation (see Fritz & MacKinnon, 2007), a sensitivity analysis revealed that collapsing over these studies ($N = 1653$) provided 80% power to detect mediation assuming correlations of 0.10 or above between X, M, and Y (Schoemann, Boulton, & Short, 2017). Thus, to examine the possibility that the effects of demographic shifts on perceptual race thresholds are attributable to greater white status threat, we conducted a bootstrapping mediation analysis (Shrout & Bolger, 2002) on our collapsed data, using the mediate function of the "Mediation" R package with 10,000 iterations. As expected, this analysis yielded a significant indirect effect such that white status threat mediated the effect of demographic shifts on thresholds for categorization of a face as a minority (A x B cross product = −0.02, $SE = 0.002$, 95% CI [−0.04, −0.01], $p < .05$.

Following recent concerns about the use of a single index of mediation and resulting Type I error inflation (Yzerbyt, Muller, Batailler, & Judd, 2018), we also used a "component" approach to provide convergent evidence for indirect mediation using the JSmediation R package. Specifically, we found that the a-path was significant (a point estimate = 0.40, $SE = 0.05$, $t = 8.30$, $p < .001$) the b-path was significant (point estimate = −0.05, $SE = 0.02$, $t = 2.06$, $p = .04$), and our indirect effect was significant (point estimate = −0.02, 95% CI [−0.04, −0.002], 5000 Monte Carlo iterations), corroborating the findings from the bootstrapping analysis.

7.3. Meta-analysis of non-white participants

Mediation cannot establish causation itself (e.g., Fiedler, Harris, & Schott, 2018) and a number of factors likely contribute to the effect of demographic change on race perception. However, the present results and theorizing suggest that demographic changes induce a threat to white status which white perceivers then might try to restore by "preserving" whiteness through perceptual hypodescent. To bolster this account, we performed exploratory analyses of the responses we recorded from non-white participants (omitted from the samples above) who presumably would not be motivated to resolve threats to white status resulting from minority advancement. We found that among our non-white participants ($N = 171$), reading about the coming demographic change increased the belief that minority gains threaten white status, such that non-white participants were more likely to see white status as threatened in a majority-minority U.S. ($n = 90$, $M = 45.08$, $SD = 29.01$) compared to control ($n = 81$, $M = 31.29$, $SD = 26.97$), $t(169) = -3.21$, $p = .002$, Cohen's $d = 0.49$, 95%CI = [0.18, 0.80]. However, changing U. S. demographics *did not affect* non-white participants' race categorization thresholds, $t(169) = 0.47$, $p = .64$, Cohen's d = −0.07, 95%CI = [−0.37, 0.23], $BF_{01} = 5.47$, suggesting "substantial evidence" for the null hypothesis (Wagenmakers, Wetzels, Borsboom, & van der Maas, 2011) that PSE does not differ by threat condition for non-white participants. Presumably this is because non-white participants were not motivated to reduce the threat to white status they perceived by altering their perception of whiteness. Therefore, these results provide tentative convergent evidence that status threat uniquely drives white decision makers to shore up their boundaries of whitness. However, future research should aim to replicate the present effects with larger samples and should also consider that "minorities" are not a monolith and could show different patterns of results based on their group's status in the racial hierarchy, contribution to changing U.S. demographics, whether they have white ancestry or not, and other experience-based or motivational factors. Indeed, we see great value in decentering whiteness in investigations of race perception and self-identification (Garay & Remedios, 2021) and hope that future research will take care to examine how demographic changes affect these processes in monoracial minority and multiracial individuals and will examine perception of Americans who themselves identify as multiracial (e.g., from the American Multiracial Faces Database; Chen et al., 2021).

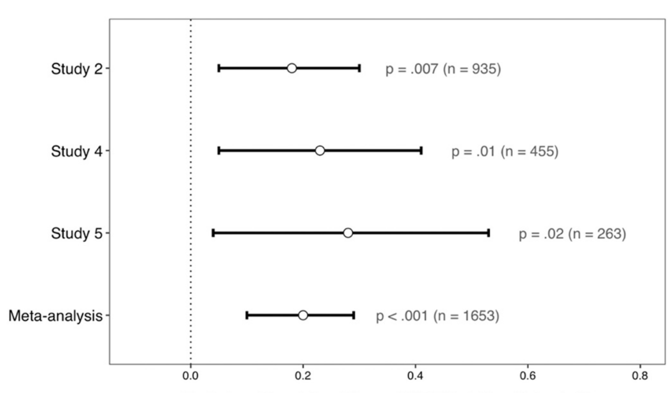

Fig. 6. Forest plot of effect sizes of Majority-Minority manipulation on PSE, separately and collapsed across Studies 2, 4, and 5. In Study 2, participants saw morphed Latino/white faces and categorized them as "Latino" or "White." In Study 4, participants saw morphed Black/white faces and categorized them as "Black" or "White." In Study 5, participants saw morphed Black/white faces and categorized them as "Not White" or "White." Meta-analysis effect sizes were based on a fixed-effects meta-analysis model given the small number of studies.

8. General discussion

In five studies, we found that threats to white status from demographic shifts altered white perceivers' categorization of mixed-race individuals as minorities. The perceptual threshold for seeing morphed Latino-white faces as minorities was lower for individuals who naturally experience more white status threat (Study 1) and for those who read about the coming demographics shifts (which induced greater white status threat; Study 2). Our findings were extended in Study 3 in which we found the same effect for real (non-morphed) racially- ambiguous Latino and white faces. Studies 4 and 5 provide supporting evidence that status threat motivates white people to raise the threshold for white group membership. Participants in those studies had a lower threshold for categorizing mixed Black-white faces as Black (Study 4) and "Not-White" (Study 5) when experiencing status threat, even though there was no mention in either study that the number of Black Americans was expected to change. Further, a high-powered mediation analysis ($N = 1653$) suggested that changing U.S. demographics induced greater white status threat which may have reduced white participants' perceptual threshold for seeing mixed-race faces as minorities. Finally, an exploratory analysis revealed our non-white participants showed no effect of threat condition on perceptual threshold for seeing mixed-race faces as minorities.

Together, these findings provide strong convergent evidence of an effect of demographic change on race perception through increased white status threat. However, it remained possible that inducing greater belief in a coming U.S. "majority-minority" demographic altered base-rate assumptions about the proportion of non-white and white individuals in the population which changed participants' categorizations. Though our demographic change manipulation explicitly described a racial minority population currently at 35% percent while poling estimates average public perception of a U.S. population that is 33% Black and 29% Hispanic alone (Gallup, 2001), we conducted an additional study to formally rule out this base rate alternative (see "Base Rate Study" in the supplement). Specifically, white-identified American participants ($N = 238$) read the geographic mobility or demographic change newspaper articles, answered questions about them, and reported demographics (as in Studies 2–5). Participants then gave their best guess about the current racial demographic make-up of the U.S. (assigning percentages to the following census-defined groups: Black or African American, White, Asian, Hispanic, Native Hawaiian or Other Pacific Islander, American Indian or Alaskan Native, and Other).

Notably, we found that the estimated proportion of minorities in the U.S. was not significantly different in the threat vs. control condition, $t(215) = -1.57$, $p = .12$, $BF_{01} = 4.27$. Descriptively, the estimated proportion of minorities in the U.S. was actually *lower* in the threat condition (53.12%) compared with the control condition (55.86%). Conversely, the estimated proportion of white people in the U.S. was descriptively higher in the threat condition (46.88%) compared with the control condition (44.14%). These results run counter to the alternative explanation that the effects of demographic threat on race perception were driven by a shifted baseline in which participants believed there were more racial minorities in the U.S. and they further bolster the present status threat account.

The fact that white perceivers raise the bar for seeing mixed-race individuals as white when threatened with numeric decline (relative to other groups) may strike some readers as counter intuitive. One could expect white perceivers to *lower* their threshold for categorizing mixed-race faces as white if they were primarily concerned with waning numbers. Indeed, previous research suggests the threat of losing a demographic majority is related to white people's desire for greater minority assimilation to white norms (Danbold & Huo, 2015). However, our results suggest that white perceivers are instead motivated to shore up the white group by visually placing more mixed-race individuals into distinctly *not white* categories, using hypodescendant categorization to protect their position as the dominant racial group (e.g., Ho et al., 2017; Krosch, Berntsen, Amodio, Jost and Van Bavel, 2013). Future research should investigate the boundary conditions, examining whether white perceivers exhibit this tendency only for groups low in the U.S. racial hierarchy (e.g., Latino and Black Americans) or whether they do so for higher status minority groups as well (e.g., East Asian Americans).

The two studies most similar to the current work (Abascal, 2020; Cooley et al., 2018) propose white status threat as the process through which demographic shifts alter race categorization but do not explicitly test this. At present, we found that across studies, status threat mediated the effect of demographic condition on race categorization threshold. Although our measured mediation cannot fully establish causation, we were able to rule out numerous alternative mediation candidates (e.g., different types of threat, perceived demographic base line changes) and demonstrate that the demographic manipulation failed to affect race perception among non-white participants. Together, this helps strengthen the role of white status threat as the process through which demographic change affects race perception. However, future research should continue to probe additional mediators of this relationship. For example, existential threat also results from demographic change (Bai & Federico, 2021) and can mediate its effect on racial outcomes (Bai & Federico, 2019), though this effect was specific to *absolute* numeric white decline. This raises the intriguing possibility that existential threat and status threat might both operate on racial boundary perception, but in an *opposing* fashion. Indeed, we predict that demographic shifts that invoke *relative* white losses may motivate white perceivers to see more faces as *non-white* to bolster white status (as we show in the present studies), while demographic shifts that invoke *absolute* white losses may motivate white perceivers to see more faces as *white* to bolster their group against extinction. Examining this interesting potential dissociation will be especially important given the very recent reporting on the 2020 census data that points to declining absolute white numbers in the US for the first time (e.g., "*Census Benchmark for White Americans: More Deaths Than Births").* Although we examined uncertainty about the future as a potential alternative mediator, future research could also more specifically examine the role of uncertainty about the white group's place in the world given connections to group identity, entitativity, and exclusion (Hogg, 2007). Future research should also examine alternative methods of manipulating demographic threat to further probe alternative mediator candidates as they might have stronger effects on prototypicality, realistic, and diversity threat (Danbold & Huo, 2015; Outten et al., 2018) as well as additional moderators of the effect of demographic threat on race perception such as beliefs about the way the economy works (e.g., Perkins, Dils, & Flusberg, 2020). Further, recent research highlights the critical need to examine interactions between sociopolitical motives and threat (i.e., the Sociopolitical Motive x Intergroup Threat Model; Ho, Kteily, & Chen, 2020) and that this model can help us adjudicate between theories that propose threat effects on categorization. Indeed, the present finding that demographic change increases categorization of mixed-race individuals as minorities among white perceivers but not non-white perceivers lends support to the assertion that demographic change threatens white status at the top rather than general threats to the racial hierarchy.

Previous work also suggests demographic change might alter perception of mixed-race faces as minorities but does not rule out that their demonstrated effects were driven by general shifts in response tendencies rather than shifts in perception. Using a tool from psychophysics, we show in Studies 2, 4, and 5 that threatening white perceivers' status as the dominant racial group via demographic change alters the way they see mixed-race individuals and lowers their perceptual threshold for seeing faces as minorities. Further, the findings of Study 3 reinforce this perceptual account – if demographic shifts induced a general response tendency to categorize a face as a minority, we would expect to see this effect across the maximally ambiguous *and* less ambiguous faces. Failing to see this suggests our effects are more likely driven by a reduction in the threshold for categorizing mixed-race faces as minority. Though these studies demonstrate how top-down influences of demographic shifts can shape the interpretation of bottom-up visual cues in race perception, future research should examine where these influences occur in the processing stream (i.e., in early visual perception within a few hundred milliseconds of face processing or more downstream cognitive processes).

The present theorizing hinged on the notion that white perceivers are motivated to shore up the white group by seeing mixed-race faces as minorities. In a supplemental experiment we sought to more directly test the role of motivated perception in categorization of mixed-race faces under demographic change (see "Census Study" in the supplement). Specifically, we ran a study identical to Study 2 with the addition of a third condition which invoked demographic threat but then implied that white status would be bolstered if more people were seen as white in the new census. That is, we added information to the threat condition meant to motivate white decision makers under demographic threat to include more mixed race faces *as white* to bolster their status. As expected, participants in both threat conditions reported greater status threat than participants in the control condition. Replicating Study 2, participants in the threat condition showed lower minority perception thresholds than those in the control condition. However, participants in the new threat + census condition did not differ from control – suggesting that when white participants are threatened but motivated to see mixed-race faces as white to preserve their status in the racial hierarchy, they reduce their threshold for seeing faces *as white*. Future research should continue to probe the role of motivation in these effects.

Importantly, this work adds to the growing body of research that suggests basic social cognitive and perceptual processes can be influenced by motivational concerns (e.g., Freeman & Ambady, 2011; Krosch & Amodio, 2014, 2019) and that threat may be an especially potent concern that can shift racial category perception (e.g., Chang et al., 2016; Gaither et al., 2016; Ho et al., 2013; Krosch & Amodio, 2014; Miller et al., 2010; Rodeheffer et al., 2012). It also extends the study of perceptual hypodescent to Latino-white mixed-race faces, a phenomenon well-documented for mixed Black-white individuals (e.g., Chen & Hamilton, 2012; Halberstadt, Sherman, & Sherman, 2010; Ho, Roberts, & Gelman, 2015; Krosch, Berntsen, Amodio, Jost and Van Bavel, 2013; Krosch & Amodio, 2014; Peery & Bodenhausen, 2008) but with mixed findings in the sparse existing literature on perceptions of mixed Latino-white individuals (Young et al., 2020).

In practical terms, our average meta-analysis effect size of $d = 0.24$ means that 60% of white Americans exposed to articles on demographic change in the U.S. should have a stricter threshold for seeing "whiteness" than the average threshold of those who did not read such an article (Cohen's U_3; Magnusson, 2021). Although this difference may appear small, when considered across the U.S. adult population, this could lead to substantial effects in the way white Americans perceive race. Back-of-envelope calculations assuming 169 million total adult newspaper readers (Nielson, 2019) and a 57.8% white population (U.S. Census Bureau, 2018) would mean about 98 million white adult newspaper readers. If each has an approximate 50% chance of reading a demographically threatening article, our effect sizes suggest an additional 29 million white Americans with stricter racial thresholds than the average of unexposed white American. Coupled with an uptick in multiracial representation in the media (Harrison, Thomas, & Cross, 2017), these threat effects on perceptual hypodescent could lead to feedback loop in which white media consumers threatened with changing demographics see more multiracial individuals as minorities, confirming demographic threats. Of course, white American news consumers are not randomly assigned to read articles in the real world and several factors likely determine exposure to such articles (e.g., access to news, interest in demographics, etc.). Nevertheless, it is important to consider the ways these small shifts in the perception of race could play out in the real world.

Recently, some have argued that the U.S. Census Bureau and the media should stop reporting such narratives that "nonwhite people will soon outnumber white people [because it] is not only divisive, but also false" (Alba, Levy, & Myers, 2021) and that an alternative U.S. demographic classification in which people with at least one white parent are classified as white could assuage some of the threat white people face (Alba, 2018). Indeed, inducing the belief in an enduring white majority or enduring white supremacy (despite changing demographics) seems to mitigate threat to white perceivers (Craig & Richeson, 2014b; Myers & Levy, 2018). However, it is clear from the present data that white perceivers are highly unlikely to classify mixed-race individuals as white, especially under perceived threat: Evidence of hypodescent was present in all studies across all stimulus and response types and was present in the control condition and made stronger in the threat condition. Indeed, white perceivers required faces to contain 60% white content to be classified as white in the control condition and 64% when threatened. Instead of embracing more inclusive notions of whiteness, white perceivers who believe white numbers are falling are less likely to see people as white. Further, research shows mixed-race individuals are very unlikely to *self*-identify monoracially as white (Rockquemore & Brunsma, 2007; Townsend, Fryberg, Wilkins, & Markus, 2012). This suggests that attempts to create more inclusive U.S. Census Bureau definitions of "white" could face resistance from both threatened monoracial white individuals as well as biracial individuals who do not find the label "white" fitting for themselves. Future research should test whether such narratives reduce or reify perceptual hypodescent.

Finally, this work adds to the growing body of research that suggests changing U.S. demographics could have negative consequences for race relations (for a review see Craig, Rucker, & Richeson, 2018) shifting who is vulnerable to discrimination. Past work finds that white decision makers experience more anger and fear toward minorities, express more explicit and implicit anti-outgroup attitudes, and show greater support for anti-minority policies when their demographic numbers are threatened (Craig & Richeson, 2014a, 2014b; Outten et al., 2012). Our findings suggest that the *number of people* likely affected by such discrimination will also increase as a result of this change. Categorization as a racial minority—and the possession of more minority features—directly impacts the degree of discrimination an individual faces (e.g., Bertrand & Mullainathan, 2003; Eberhardt et al., 2006; Hebl et al., 2012; Maddox, 2004). Given the ubiquity of news headlines describing demographic shifts in the U.S. (e.g., "Whites no longer a majority in U.S. by 2043"), many white Americans are likely to experience threats to their status, which can lead to greater anti-minority attitudes and behaviors. As the number of people who identify as more than one race continues to increase to record levels (U.S. Census Bureau, 2018), amplified discriminatory behavior could affect a rapidly increasing number of people as the U.S. inches closer to becoming a majority-minority nation. By examining how demographic shifts alter *who* is perceived as a minority in the first place, we aim to identify those most vulnerable to the consequences of demographic change.

Data availability

We report all data exclusions, manipulations, conditions, and measures. Studies 2, 4, and 5 were preregistered. Preregistrations, data files, analysis scripts, and stimuli are available here: https://osf.io/7vfbn/.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jesp.2021.104266.

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SUPPLEMENTAL MEASURES AND MATERIALS

Manipulation and categorization task cover story

Participants were first told after their consenting:

> *For the first part of this study, you will be reading a newspaper article about recent events in the news and answering questions about the article to help us pilot stimuli for an upcoming task.*

After participants finished these comprehension questions, they were told:

> *Thank you for helping pilot the newspaper stimuli. Next, you will help us pilot stimuli for a separate study by completing a face categorization task.*

On the next slide (which was presented on an html page with different formatting from the previous portion of the study which was presented in Qualtrics), participants were told:

> *PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY.*
>
> *In this task, you will see a series of faces. You are to identify each face as either **black** or **white**. If the face is black, press the "Q" key. If the face is white, press the "P" key. When you are ready to start the experiment, please press 'Begin'.*

Participants were not given a response deadline when making their categorizations.

Threat condition article (all experiments)

In a Generation, Racial Minorities May Be the U.S. Majority

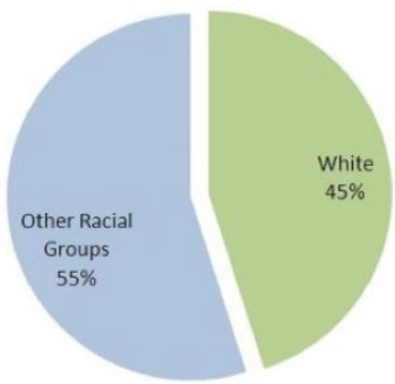
By: S. Roberts

A new analysis of 2010 U.S. Census Bureau data suggests that America might become a White-minority nation faster than once predicted. The nation's racial minority population is steadily rising and currently makes up 35 percent of the United States. This advances an unmistakable trend that could make racial minorities the new American majority by midcentury, a transformation that is occurring much faster than anticipated just a few years ago.

Demographers calculate that by 2042, Americans who identify themselves as Hispanic, Black, Asian, American Indian, Native Hawaiian, or Pacific Islander will together outnumber non-Hispanic Whites. Four years ago, officials had projected the shift would come in 2050. This shift can already be seen in several U.S. states; currently, four states as well as the District of Columbia have minority populations that exceed 50 percent.

The main reasons for the accelerating change are rapid immigration growth and significantly higher birthrates among racial and ethnic minorities. As White baby boomers age past their childbearing years, younger Hispanic parents are having children – and driving U.S. population growth. New estimates show minorities increased to 107.2 million people in 2009, largely boosted by a surge in Hispanic births. During this time, the White population remained flat, making up roughly 199.9 million, or 65 percent, of the country. The group predicted to post the most dramatic gain, Hispanics, is projected to nearly triple in size by 2050. There are now roughly 9 births for every 1 death among Hispanics, compared to a roughly one-to-one ratio for Whites.

A Majority-minority US in 2042



Source: US Census Bureau

The Census Bureau projects that ethnic and racial minorities will constitute a majority of the nation's children under 18 by 2023 and of working-age Americans by 2039. Indeed, about 49.6 percent of babies born in 2011 were White, compared to about 50.4 percent who were racial minorities.

The latest figures are predicated on current and historical trends, which demographers say can be reasonably expected to continue.

"What's happening now in terms of increasing diversity is unprecedented," said Campbell Gibson, a census demographer.

Comprehension Questions

1. Which group is experiencing the fastest increase in population growth in the U.S. today?
2. By when do demographers predict that racial minorities will make the new American majority?
3. What is one of the main reasons for the accelerating change in minority population?

Control condition article (all experiments)

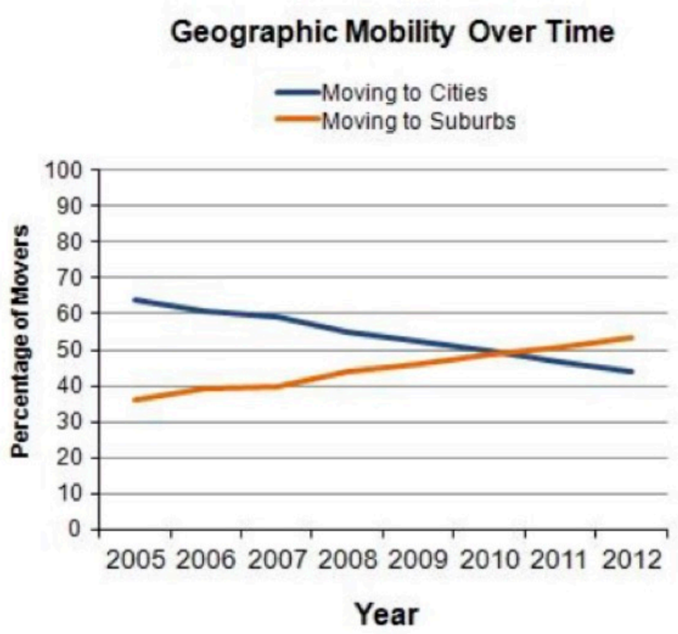
U.S. Census Bureau Reports Residents Now Move at a Higher Rate

By: S. Roberts

New U.S. Census Bureau data suggest that the rate of geographical mobility, or the number of individuals who have moved within the past year, is increasing. The national mover rate increased from 11.9 percent in 2008 (which was the lowest rate recorded since the U.S. Census Bureau began tracking the data) to 12.5 percent in 2009.

According to the new data, 37.1 million people changed residences in the United States within the past year. Eighty-four percent of all movers stayed within the same state. Renters were more than five times more likely to move than homeowners.

The new estimates also reveal that many of the nation's fastest-growing communities are suburbs. Specifically, principal cities within metropolitan areas experienced a net loss of 2.1 million movers, while the suburban areas had a net gain of 2.4 million movers. For those who moved to a different county or state, the reasons for moving varied considerably depending on the length of their move. Overall, four out of 10 (43.7 percent) moved due to housing-related reasons, such as the desire to live in a new or better home or apartment.



These geographic mobility data are used to determine the extent of mobility of the U. S. population and the resulting redistribution. Migration data are collected as a part of the Annual Social and Economic Supplement (ASEC) to the Current Population Survey (CPS). How populations change has implications for federal, state, and local governments, as well as for private industry. The latest figures are predicated on current and historical trends, which can be thrown awry by several variables, including prospective overhauls to public and economic policy.

Comprehension Questions

1. How has the number of individuals who have moved within the past year changed from 2008 to 2009?
2. What type of communities are experiencing net gains in movers?
3. What are geographic mobility data used to determine?

Alternative Mediators – Studies 2a and 4a

System threat (adapted from Jost et al., 2007):

- "The American way of life is threatened" (0 = *strongly disagree* to 100 = *strongly agree*)
- "What is your view of American society's trajectory?" (0 = *getting much worse every year* to 100 = *getting much better every year*)

System justification (Kay & Jost, 2003):

- "People usually get what they deserve in American society" (0 = *strongly disagree* to 100 = *strongly agree*)

Perceived threat to Whites' societal status (adapted from Outten et al., 2012):

- "Increases in racial minorities' status will reduce White Americans' status" (0 = *strongly disagree* to 100 = *strongly agree*)

Perceived Uncertainty (Craig & Richeson, 2014):

How certain do you feel American society's future is?

(Measured from 0 - "Extremely Uncertain" to 100 - "Extremely Certain")

Ideology I (from Craig & Richeson, 2014):

(Measured from 0 = "*Strongly Disagree"* to 100 = "Strongly Agree")

- To what degree do you endorse aspects of conservative political ideology?
- To what degree do you endorse aspects of liberal political ideology?

Ideology II:

(Measured from 0 - "Extremely Liberal" to 100 - "Extremely Consevative")

- Where on the following scale of political orientation would you place yourself (overall, in general)?
- In terms of social and cultural issues in particular, how liberal or conservative are you?
- In terms of economic issues in particular, how liberal or conservative are you?

Ideology III:

To what extent should the following be increased or decreased?

(Measured from 0 = "Increased a lot" to 100 = "Decreased a lot")

- The required time to be eligible for U.S. citizenship
- Foreign immigration to the United States
- Funding of the U.S. Department of Defense
- Minimum wage

Please indicate your degree of support for the following policies:

(Measured from 0 = "*Strongly opposed"* to 100 = "Strongly in favor")

- Affirmative Action
- English as the official U.S. language
- Health-care reform
- Same-sex marriage
- Building more prisons
- Unions
- Drilling in the Arctic National Wildlife Refuge

Zero-sum beliefs about resource distribution between races (Krosch & Amodio, 2014)
(Measured from 0 = "Strongly Disagree" to 100 = "Strongly Agree")

- When Black people make economic gains, White people lose out economically
- Black people tend to open up small businesses, which means that there are fewer business opportunities available to White people.
- Money spent on social services for Black people means less money for services for White people.
- The more power Black people obtain in America, the more difficult it is for White people to obtain power.
- As more Black people take advantage of education, there are fewer spots and opportunities available for White people.
- Black people are taking our jobs.
- Allowing Black people to decide on political issues means that White people have less say in how the country is run.
- More Black people in positions of power means fewer opportunities for White people.
- The more Black people in America, the harder it is for White people to get ahead.
- Black people have too much to say about political matters.
- Black people have been trying to get ahead economically, at the expense of White people.
- More good jobs for Black people means fewer good jobs for White people.
- Financial aid to Black people hurts White people.
- White people may no longer have a say in how the country is run because Black people are trying to take control.
- More good jobs for Black people means fewer good jobs for members of other groups.
- The more influence Black people have in local politics, the less influence members of other groups will have in local politics.
- As more good housing and neighborhoods go to Black people, the fewer good houses and neighborhoods there will be for members of other groups.
- Many Black people have been trying to get ahead economically at the expense of other groups.

Attitudes toward Blacks scale (Brigham, 1993)

(Measured from 0 = “Strongly Disagree” to 100 = “Strongly Agree”)

- I would rather not have Black people live in the same apartment building I live in.
- I get very upset when I hear a White person make a prejudicial remark about Black people.
- Black and White people are inherently equal.
- I would not mind at all if a Black family with about the same income and education as me moved in next door.
- It would not bother me if my new roommate were Black.
- Whites should support Blacks in their struggle against discrimination and segregation.
- If a Black person were put in charge of me, I would not mind taking advice and direction from him or her.
- I think that Black people look more similar to each other than White people do.
- The federal government should take decisive steps to override the injustices Black people suffer at the hands of local authorities.
- I would probably feel somewhat self-conscious dancing with a Black person in a public place.
- Interracial marriage should be discouraged to avoid the "who-am-I?" confusion which the children feel.
- Black people are demanding too much too fast in their push for equal rights.
- I enjoy a funny racial joke, even if some people may find it offensive.
- If I had a chance to introduce Black visitors to my friends and neighbors, I would be pleased to do so.
- I favor open housing laws that allow more racial integration of neighborhoods.
- Generally, Black people are not as smart as White people.
- Some Black people are so touchy about race that it is difficult to get along with them.
- It is likely that Black people will bring violence to neighborhoods when they move in.
- Racial integration (of schools, businesses, residences, etc.) has benefited both Whites and Black people.
- I worry that in the next few years I may be denied my application for a job or a promotion because of preferential treatment given to minority group members.

Ascent of Man scale (Kteily, Bruneau, Waytz, & Cotterill, 2015)

People can vary in how human-like they seem. Some people seem highly evolved whereas others seem no different than lower animals. Using the image below, indicate using the sliders how evolved you consider the average group member of each group to be:

Black People

White People

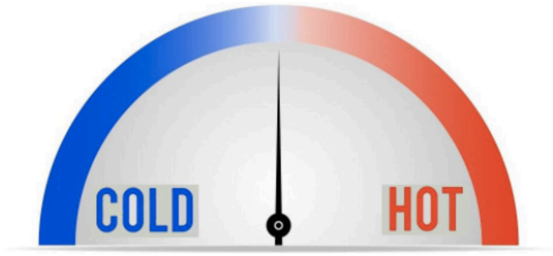
Asian People

Latino People

Feeling Thermometer (Amodio & Devine, 2006)

Below you will see something that looks like a thermometer. You will be using this to indicate your personal overall evaluation of these racial groups.

Here's how it works: if you have a favorable attitude toward members of this group, you would give the group a score somewhere between 50° and 100°, depending on how favorable your evaluation is. On the other hand, if you have an unfavorable attitude toward members of this group, you would give the group a score somewhere between 0° and 50°, depending on how unfavorable your evaluation is.



0 10 20 30 40 50 60 70 80 90 100

Black People

0 10 20 30 40 50 60 70 80 90 100

White People

Alternative Mediators – Study 3

Diversity Threat and Influence (Outten et al., 2018)

- White Americans should be worried about its place in the future of the U.S.
- White Americans should be threatened by growing ethnic diversity.
- White Americans will benefit from increasing diversity in America. (r)

(Measured from 0 - "Strongly Agree" to 100 - "Strongly Disagree")

- How much influence will ethnic minorities have over American society in the future?
- How much influence will White Americans have over American society in the future? (r)

(Measured from 0 - "Very Little" to 100 - "Very Much")

Prototypicality Threat (Danbold & Huo, 2014)

- I fear that in 40 years time, it won't be clear what it means to be American.
- I believe that there will always be a place for White people in American society. (r)
- I fear that in 40 years time, White people will not represent the American identity.

(Measured from 0 - "Strongly Disagree" to 100 - "Strongly Agree")

Realistic Threat (Stephan et al., 1999, as in Danbold & Huo, 2014)

- The growth of other ethnic groups has increased the tax burden on White Americans.
- Social services have become less available to White Americans because of the growth of other ethnic groups.
- Members of other ethnic groups are not displacing White Americans from their jobs. (r)

(Measured from 0 - "Strongly Disagree" to 100 - "Strongly Agree")

Symbolic Threat (Stephan et al., 1999, as in Danbold & Huo, 2014)

- The values and beliefs of other ethnic groups regarding moral issues are not compatible with the values and beliefs of White Americans
- The growth of other ethnic groups is undermining American culture.
- The values and beliefs of other ethnic groups regarding work are not compatible with the values and beliefs of White Americans.

(Measured from 0 - "Strongly Disagree" to 100 - "Strongly Agree")

Perceived Uncertainty (Craig & Richeson, 2014)

How certain do you feel American society's future is?

(Measured from 0 - "Extremely Uncertain" to 100 - "Extremely Certain")

System Threat (Craig & Richeson, 2014)

What is your view of American society's trajectory?

(Measured from 0 - "Getting much worse every year" to 100 - "Getting much better every year")

Outnumbered

Personally, how likely do you think it is that non-White people will outnumber White people in America by 2042? *(Measured from 0 - "Extremely Unlikely" to 100 - "Extremely Likely")*

SUPPLEMENTAL ANALYSES

Study 1

Here we examined PSE on status threat controlling for participant demographics that might be correlated with status threat. Specifically, we regressed PSE onto status threat adjusting for the participant demographics we collected (gender, income, education level, age). Group status remained a robust predictor in this model ($B = -.19$, $t = -3.49$, $p < .001$). Interestingly, participants with a doctoral degree or master's degree showed marginally less status threat than participants who finished high school or had a GED (B's > 10.75, p's > .10). Participants with a household income of > $50-59K showed greater status threat compared those who made < $30K (B's > 9.97, p's <= .05). However, these subsamples were *tiny* so these findings should be regarded as very tentative. No other effects emerged as significant predictor of threat.

Study 2

Study 2 was originally run as two separate studies. Study 2a, run on 9/13/18 as part of a student's senior honors thesis, yielded marginal effects. Participants categorized faces from an original set of 110 Latino-White stimuli. However, a closer analysis of these stimuli in a separate pilot revealed that 15 faces were categorized further than 40% from their "objective" racial content (e.g., a 20% White face that was categorized as Latino more than 60% of the time). Thus, we conducted Study 2b on 7/25/19 when we revived the project, as a preregistered high-powered replication of Study 2a with a reduced face stimulus set of 95 faces (no face bin had fewer than 8 unique faces). Adding sample (2a vs. 2b) did not improve model fit so we present results together in the main manuscript and separately here for completeness:

Study 2a Method and Results

Method. We recruited White-identified, non-Latinx Amazon Mechanical Turk workers in exchange for $2 for approximately 20 minutes of their time. 294 met our preregistered and reported inclusion criteria (mean age = 40.05, SD = 20.67; 177 identified as female and 117 identified as male). See main text for procedure and materials.

Results. We found a significant effect of condition on perceptions of group status threat, such that participants in the threat condition reported greater group-status threat (n = 150, $M = 43.35$, $SD = 29.19$) than participants in the control condition (n = 144, $M = 31.56$, $SD = 27.37$), $t(292) = 3.57$, $p < .001$, Cohen's $d = 0.42$, 95%CI = [0.18, 0.65] and a non-significant effect of condition on PSE in the expected direction, such that participants in the threat condition exhibited a lower PSE ($M = 46.26$, $SD = 12.87$) than participants in the control condition ($M = 48.23$, $SD = 14.49$), $t(292) = 1.29$, $p = .20$, Cohen's $d = -0.15$, 95%CI = [-0.38, 0.08].

We found no significant effects of the manipulation on alternative mediator candidates, including political ideology measured in scales of self-reported conservatism (Political Ideology I and II) or conservative policy support (Political Ideology III), the Attitudes toward Blacks scale, or a feeling thermometer. The manipulation did affect scores on the Ascent of Man scale, such that the threat condition *increased* humanization across all races (see Table S1).

Table S1. Alternative Mediators by Condition

	Threat Condition		Control Condition			
Threat Measure	*M*	*SD*	*M*	*SD*	*t*	*p*
Political Ideology I	45.2	33.0	40.7	28.9	1.29	.20
Political Ideology II	46.5	30.0	42.3	28.7	1.23	.22
Political Ideology III	42.3	13.8	40.1	14.1	1.60	.11
ATB	23.0	15.9	22.3	16.7	0.35	.72
ZSB	17.6	17.5	15.7	21.0	0.40	.40
AOM - Black	95.2	13.1	90.9	17.9	2.37	.02
AOM - Latino	96.0	10.1	90.9	17.8	3.00	.002
AOM - Asian	97.5	7.34	92.9	15.7	3.25	.001
AOM - White	97.7	7.51	94.4	13.2	2.62	.009

Warmth - Black	79.9	21.2	80.2	21.2	0.13	.90
Warmth - Latino	81.2	19.6	80.0	21.0	0.51	.61
Warmth - Asian	82.7	18.1	83.1	18.3	0.16	.88
Warmth - White	84.2	18.0	81.9	19.6	1.02	.31

Study 2b Method and Results

Method. We recruited White-identified, non-Latinx Amazon Mechanical Turk workers in exchange for $1 for approximately 5 minutes of their time. 641 met our preregistered inclusion criteria (mean age = 37.85, SD = 12.74; 375 identified as female, 260 identified as male, 2 identified as non-binary, and 4 did not report their gender). See main text for procedure and materials.

Results. We found a significant effect of condition on perceptions of group status threat, such that participants in the threat condition reported greater group-status threat (n = 324, M = 43.52, SD = 29.35) than participants in the control condition (n = 317, M = 32.07, SD = 28.24), $t(639) = 5.04$, $p < .001$, Cohen's d = 0.40, 95%CI = [0.24, 0.56]. A significant effect of condition on PSE, such that participants in the threat condition exhibited a lower PSE (M = 45.86, SD = 13.67) than participants in the control condition (M = 48.51, SD = 14.80), $t(639)$ = 2.36, p = .02, Cohen's d = -0.19, 95%CI = [-0.34, -0.03].

Study 2 Analysis with Only Preregistered Exclusions

Excluding participants based on *only* our preregistered and exclusion criteria yielded data from 94 participants (eight more than reported in the main text). Participants in the threat condition reported greater threat to White status (n = 475, M = 43.51, SD = 29.25) than those in the control condition (n = 468, M = 32.00, SD = 27.96), $t(941) = 6.18$, $p < .001$, Cohen's d = 0.40, 95%CI = [0.27, 0.53]. As expected, there was also a significant effect of condition such that participants in the threat condition exhibited a lower PSE (M = 45.88, SD = 13.56) than those in the control condition (M = 49.05, SD = 15.64), $t(934) = 3.31$, $p < .001$, Cohen's d = 0.22, 95%CI = [0.09, 0.34].

Study 3

Table S2. Alternative Mediators by Condition

	Threat Condition		Control Condition			
Threat Measure	*M*	*SD*	*M*	*SD*	*t*	*p*
Diversity Threat	3.88	0.65	3.78	0.72	0.43	.67
Prototypicality Threat	2.73	1.37	2.61	1.38	0.60	.55
Realistic Threat	2.76	1.55	2.65	1.62	0.48	.63
Symbolic Threat	2.37	1.50	2.22	1.45	0.68	.50
Uncertainty Threat	4.18	1.63	4.08	1.77	0.40	.69
System Threat	3.58	1.41	3.73	1.44	0.75	.46
Likelihood of Maj-Min US	5.65	1.22	4.86	1.56	3.89	< .001

Study 4

Study 4 was originally run as two separate studies: Study 4a, a small-N pilot (run on 4/11/18), yielding non-significant results in the expected direction and our preregistered replication study (S4b, run on 4/18/18), which yielded significant results. S4a was identical to S4b except that it included the alternative mediator candidates from Study 2a which again did not show significant effects of the manipulation, so we omitted them from Study 4b. Adding a sample variable (S4a vs. S4b) and its interaction with the manipulation to the model did not improve fit, $F = 0.21$, $p = 0.81$, nor was there a significant main effect of sample or interaction of sample and manipulation on PSE (p's > .62). Thus, we feel confident merging the samples and present the combined results in the main manuscript and separately here for completeness:

Study 4a

Method. We recruited White-identified, non-Latinx Amazon Mechanical Turk workers in exchange for $2 for approximately 20 minutes of their time. 181 met our reported inclusion criteria (mean age = 38.32, SD = 12.77; 98 identified as female and 82 identified as male and 3 did not report their gender). See main text for procedure and materials.

Results. We found a significant effect of condition on perceptions of group status threat, such that participants in the threat condition reported greater group-status threat (n = 94, M = 48.30, SD = 32.63) than participants in the control condition (n = 87, M = 33.18, SD = 29.72), $t(179) = 3.31$, $p = .001$, Cohen's $d = 0.49$, 95%CI = [0.19, 0.79] and a non-significant effect of condition on PSE in the expected direction, such that participants in the threat condition exhibited a lower PSE (M = 45.02, SD = 10.74) than participants in the control condition (M = 47.70, SD = 10.90), $t(179) = 1.66$, $p = .10$, Cohen's $d = -0.25$, 95%CI = [-0.54, 0.05].

Table S3. Alternative Mediators by Condition

	Threat Condition		Control Condition			
Threat Measure	*M*	*SD*	*M*	*SD*	*t*	*p*
Political Ideology I	40.4	32.3	36.2	29.7	0.90	.37
Political Ideology II	44.0	31.1	37.9	27.9	1.39	.17
Political Ideology III	39.7	15.7	37.7	12.4	0.94	.35
System Threat	43.2	16.6	43.2	16.5	0.02	.99
System Justification	41.1	25.5	46.73	26.9	1.45	.15
Certainty	43.8	26.1	46.6	26.2	0.74	.46
ATB	25.5	20.4	25.3	18.1	0.07	.94
ZSB	21.6	22.7	17.1	21.8	1.37	.17

AOM - Black	90.8	21.7	92.7	18.5	0.61	.54
AOM - Latino	92.6	18.3	93.3	16.4	0.25	.80
AOM - Asian	95.5	14.7	95.4	13.4	0.01	.99
AOM - White	97.4	7.14	96.5	11.1	0.62	.54
Warmth - Black	74.6	27.9	74.1	26.8	0.12	.90
Warmth - Latino	76.9	23.7	77.2	21.3	0.10	.92
Warmth - Asian	80.8	19.9	79.8	20.3	0.32	.75
Warmth - White	83.0	19.8	80.1	21.4	0.94	.35

Study 4b

Method. We recruited White-identified, non-Latinx Amazon Mechanical Turk workers in exchange for $1 for approximately 5 minutes of their time. 276 met our preregistered inclusion criteria (mean age = 38.26, SD = 12.53; 161 identified as female, 115 identified as male. See main text for procedure and materials.

Results. We found a significant effect of condition on perceptions of group status threat, such that participants in the threat condition reported greater group-status threat (n = 138, M = 41.75, SD = 28.76) than participants in the control condition (n = 138, M = 31.62, SD = 27.93), $t(274) = 2.97$, $p = .003$, Cohen's d = 0.36, 95%CI = [0.12, 0.60]. A significant effect of condition on PSE, such that participants in the threat condition exhibited a lower PSE (M = 44.16, SD = 12.94) than participants in the control condition (M = 47.05, SD = 10.29), $t(274)$ = 2.05, $p = .04$, Cohen's d = -0.25, 95%CI = [-0.49, -0.01].

Study 5 Analysis with Only Preregistered Exclusions

Excluding participants based on *only* our preregistered and exclusion criteria yielded data from 264 participants (one more than reported in the main text). Participants in the threat condition reported greater threat to White status (n = 21, M = 48.14, SD = 31.55) than those in the control condition (n = 132, M = 35.16, SD = 28.82), $t(262) = 3.49$ $p < .001$, Cohen's d = 0.43, 95%CI = [0.18, 0.68]. As expected, there was also a significant effect of condition such that participants in the threat condition exhibited a lower PSE (M = 35.51, SD = 14.44) than those in the control condition (M = 39.51, SD = 14.44), $t(934) = 2.45$, $p = .01$, Cohen's d = 0.30, 95%CI = [0.06, 0.55].

Data Across Experiments - Face Stimuli and Response Option Moderators

Exploratory tests of threat condition on PSE as a function of stimulus and response type revealed the effect of condition on PSE was significant across studies, $F(1,1647) = 17.33$, $p < .001$ replicating the mini meta-analysis. Further, the main effect of study was significant, $F(2,1647) = 58.03$, $p < .001$, such that PSE was lowest for the White/Not White categorization of mixed Black-White faces (M = 37.35, SD = 13.19; Study 5), followed by White/Black categorizations of mixed Black-White faces (M = 45.92, SD = 11.30; Study 4) followed by White/Latino categorizations of mixed Latino-White faces (M = 47.20, SD = 14.10; Study 2). Not White categorization thresholds for mixed Black-White faces (Study 5) was significantly lower than Black categorization thresholds for mixed Black-White faces (Study 4) and Latino categorization thresholds for mixed Latino-White faces (Study 2), p's < .001. Black categorization thresholds for mixed Black-White faces were marginally lower than Latino categorization thresholds for mixed Latino-White faces, $p = .09$. This suggests that participants engaged in greater hypodescent for Black-White vs. Latino-White stimuli and the greatest hypodescent when given a "Not White" response option. Although the

Condition x Study interaction was non-significant, $F(2,1647) = 0.23$, $p = .79$, the effect of the status threat manipulation on PSE in Study 5 (d = .28) was descriptively larger than in Study 4 (d = .23) which used identical stimuli but “Black” vs. “White” categorization (d = .23). This suggests that White perceivers may draw more exclusive boundary around Whiteness when they have greater flexibility to categorize mixed race faces as non-White.

Base Rate Study

It is possible the demographic change article could have shifted base rate assumptions about the demographic breakdown in the U.S. such that participants in the threat condition thought a larger proportion of Americans are non-White compared with participants in the control condition, and thus would categorize more faces as minorities or have a lower threshold for seeing faces as minority. In order to test this alternative, we collected a new group of white-identified American participants (N = 238) who read about geographic mobility or demographic change in short newspaper articles and answered questions about them (as in Studies 2-5) in order to ostensibly help us pilot stimuli for an upcoming study. We then asked participants the demographic questions from Studies 2-5 as well as their best guess about the current racial demographic make-up of the U.S. (assigning percentages to the following census-defined groups: Black, White, Asian, Hispanic, PAC, American Indian, and Other). Filtering out non-conscientious responders who said any of the four largest racial group (Black, White, Asian, Hispanic) was 100% or 0% of the U.S. population left us with N = 217.

Notably, we found that estimated proportion of minorities in the U.S. was not significantly different in the threat vs. control condition, $t(215) = -1.57$, $p = 0.12$, $BF_{01} = 4.27$. This suggests the data were twice as likely to have occurred under the null hypothesis that base rate estimates are the same in the control vs. threat condition compared to the alternative hypothesis that base rate estimates of the minority population are higher in the threat than control condition. Further, the estimated proportion of minorities in the U.S. was descriptively *lower* in the threat condition (53.12%) compared with the control condition (55.86%) while the estimated White population was higher in the threat (46.88%) compared with the control condition (44.14%). These results run counter to the alternative explanation that the effects of demographic threat on race perception were driven by a shifted baseline in which participants believed there were more racial minorities in the U.S.

Examining specific minority groups, we found that in the threat condition participants estimated a greater proportion of the U.S. to be Hispanic (17%) compared with the control condition (13.87%), $t(215) = -3.53$, $p < .001$. While this could support the alternative shifted baseline account for the Latino/White categorization studies, the opposite was true for estimates of the Black population: Participants in the threat condition reported a lower estimate (17.87%) compared with control (21.83%), $t(215) = 3.93$, $p < .001$ (see below for a full table). Thus, it is hard to see how this baseline account could provide an alternative explanation for our current data across studies.

	Control (n = 109)	**Threat (n = 108)**	**t-value**	**p-value**
White	44.14% (13.96)	46.88% (11.67)	-1.57	0.12
Black	21.83% (8.46)	17.87% (6.07)	3.96	< 0.001
Hispanic	13.87% (6.43)	16.98% (6.54)	-3.53	< 0.001
Asian	5.02% (3.05)	4.10% (3.22)	1.53	0.13
PAC	3.04% (2.60)	3.22% (2.82)	-0.47	0.64
Amer Indian	5.25% (3.94)	4.62% (3.48)	1.26	0.21

Other	2.63% (3.21)	2.16% (2.60)	1.18	0.24

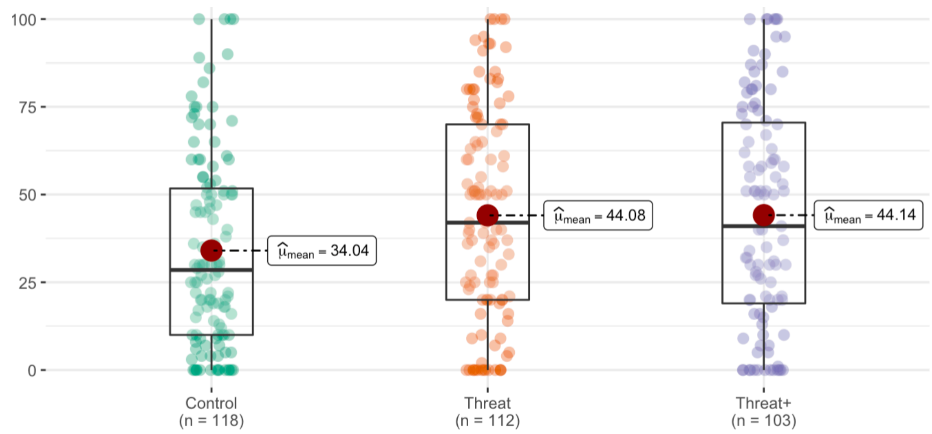
Census Study

To further test the role of motivated perception in categorization of mixed-race faces under demographic change, we added a new condition which implied that White status would be bolstered if more people were seen as White in the new census. That is, we added information to the threat condition meant to motivate White decision makers under demographic threat to *include* more mixed race faces *as white* in order to bolster a numeric majority.

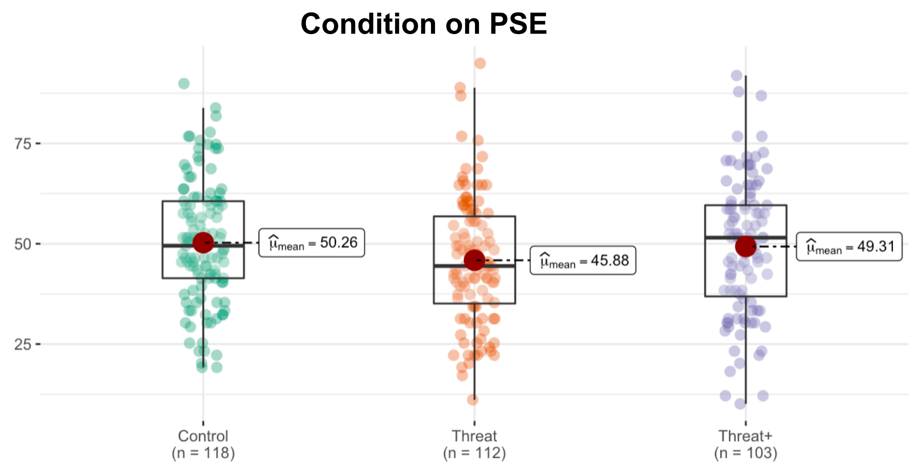
Method. We recruited white-identified, non-Latinx Amazon Mechanical Turk workers in exchange for $1 for approximately 5 minutes of their time. 336 met our preregistered inclusion criteria (193 female, 142 male, 1 transman, M_{age} = 37.53, SD_{age} = 11.36). Materials and procedure were identical to Study 2 except that in addition to a threat and control condition, we included a new threat + census study in which participants read the following information after the article about demographic threat: *“Some have argued that in order to maintain Whites’ place as the dominant group in America, the 2020 census must reflect that White Americans outnumber minorities.”*

Results. We conducted one-way ANOVAs to determine the effect of condition on white status threat and PSE, followed by Tukey HSD tests. We found a significant main effect of condition on white status threat, $F(2,330) = 4.38$, $p = .01$, such that participants in the control condition reported less white status threat (M = 34.04, SD = 15.22) than those in the threat (M = 44.08 SD = 15.96) or threat+census conditions (M = 44.14, SD = 16.98), p’s < .03. The threat and threat+census conditions did not differ on group status threat, $p = .89$. Contrast analyses [control = -1, threat = .5, threat+census = .5] confirm that participants in the threat and threat+census conditions showed significantly greater status threat than participants in the control condition, $F(1,330) = 8.76$, $p = .003$.

Condition on Status Threat



Further, we found a marginal main effect of condition on PSE, $F(2, 330) = 2.34$, $p = .10$, such that participants in the control condition reported a higher PSE ($M = 50.26$, $SD = 15.22$) than those in the threat condition ($M = 45.88$, $SD = 15.96$), $p < .10$, as in our other studies. Participants in the threat+census conditions ($M = 49.31$, $SD = 16.98$) did not differ from either control or threat, p's > .26, but descriptively had a PSE more similar to the *control* condition than the threat condition. Contrast analyses [control = -.5, threat = 1, threat+census = -.5] confirm that participants in the threat condition control showed significantly lower PSE than participants in the threat+census conditions, $F(1,330) = 4.49$, $p = .03$.



This suggests that although participants found demographic change threatening in both the threat and threat+census condition, they had two different behavioral uses for face categorization to try to resolve that threat. In the threat condition, as our other studies, participants adopted a lower threshold for seeing faces as Black. When threatened and motivated to see faces as white, they showed a lower threshold for seeing faces as White.